

MANDAKH, Mongolia

WMO: 443480

Lat: 44.4014N

Lon: 108.2492E

Elev: 4288

StdP: 12.56

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-11.2	-23.6	1.7	-8.6	-19.4	2.2	-4.9	35.1	4.8	31.5	0.5	14.8	320	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	89.9	64.2	86.6	62.9	83.6	61.8	69.7	83.6	67.2	80.4	65.2	78.3	11.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	108.8	77.9	62.6	99.6	75.4	60.4	92.1	72.5	36.7	83.1	34.4	80.3	32.6	78.0	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.8	27.5	23.4	DB	-19.6	95.1	5.2	3.0	-23.4	97.3	-26.4	99.1	-29.4	100.8	-33.2	103.0	(4)
(5)			WB	-19.8	71.1	5.2	4.7	-23.6	74.5	-26.7	77.3	-29.6	79.9	-33.4	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	4.3	11.3	27.9	43.5	55.5	67.1	72.5	69.1	57.8	41.6	21.8	8.0		
	DBStd	25.38	8.88	10.56	10.73	9.74	9.02	6.73	5.52	6.26	8.15	9.24	10.16	8.47		
	HDD50	5938	1417	1084	687	240	48	0	0	0	28	287	846	1301		
	HDD65	9612	1882	1504	1151	646	315	54	7	29	237	725	1296	1766		
	CDD50	2352	0	0	1	44	218	512	696	593	261	27	0	0		
	CDD65	551	0	0	0	0	20	116	238	157	20	0	0	0		
	CDH74	5111	0	0	0	13	276	1142	2157	1353	167	3	0	0		
	CDH80	1584	0	0	0	1	56	355	740	417	15	0	0	0		
(14) Wind	WSAvg	11.4	12.2	11.5	12.2	12.9	13.7	10.7	9.6	9.7	10.1	10.8	11.0	11.9		
(15) Precipitation	PrecAvg	5.1	0.1	0.1	0.2	0.1	0.4	0.8	1.3	1.2	0.5	0.3	0.2	0.1		
	PrecMax	8.5	0.2	0.5	0.6	0.4	1.7	1.4	3.4	3.1	1.4	0.9	0.5	0.3		
	PrecMin	3.1	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.3	0.0	0.1	0.2	0.1	0.4	0.4	0.6	0.8	0.4	0.2	0.1	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.4	41.9	62.0	75.2	86.4	92.7	95.5	93.0	82.1	71.1	51.3	32.7		
		MCWB	23.4	32.3	42.1	50.9	56.5	66.2	70.8	66.9	57.8	51.6	38.9	26.1		
	2%	DB	25.0	36.1	54.7	69.9	81.1	88.0	90.4	88.2	78.7	65.3	45.4	27.7		
		MCWB	19.0	28.4	38.7	47.9	54.9	60.6	65.6	65.2	56.6	48.9	35.4	22.6		
	5%	DB	20.7	31.6	49.8	64.7	76.9	84.0	87.1	84.5	75.5	60.5	40.4	23.6		
		MCWB	16.6	25.1	36.3	45.3	53.1	60.2	63.8	63.2	55.7	45.8	32.4	19.5		
	10%	DB	16.9	27.5	44.7	60.2	72.9	80.4	84.1	81.3	71.9	56.0	36.4	20.2		
		MCWB	13.8	22.3	33.0	43.5	51.5	59.2	63.4	61.5	54.4	43.1	29.5	16.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	33.4	43.0	54.6	60.9	70.7	75.6	71.2	63.7	54.4	39.6	26.7		
		MCDB	30.9	40.8	58.4	71.1	79.1	87.6	92.8	86.0	76.8	68.7	50.3	32.6		
	2%	WB	19.6	28.7	39.6	49.7	57.4	66.3	70.6	68.7	60.2	50.3	36.1	22.8		
		MCDB	24.6	35.9	53.4	66.1	74.4	80.3	84.1	83.0	73.7	63.5	45.1	27.4		
	5%	WB	16.6	25.5	36.6	46.5	54.8	64.0	68.2	66.1	57.8	46.5	33.0	19.7		
		MCDB	20.4	31.4	49.4	63.0	72.1	77.8	80.7	80.3	70.6	58.1	40.5	23.4		
	10%	WB	13.9	22.3	33.3	43.8	52.7	61.6	66.2	63.9	55.7	43.6	30.2	16.9		
		MCDB	16.9	27.4	44.4	59.3	70.6	75.8	78.5	77.4	68.5	55.4	36.7	20.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	18.0	20.9	20.9	22.0	19.9	18.3	18.5	19.4	18.3	16.1	13.8		
		MCDBR	19.7	23.1	25.9	26.8	26.7	23.8	22.1	21.6	23.3	23.1	20.2	17.4		
		MCWBR	15.6	17.2	16.3	15.7	13.9	11.2	9.3	10.1	11.5	13.7	14.0	13.4		
		MCDBR	19.3	22.8	25.4	25.5	24.8	20.8	18.7	19.1	19.5	21.9	19.7	17.3		
	5% WB	MCWBR	15.5	17.2	16.5	16.1	14.6	11.5	10.5	10.5	11.9	14.3	14.1	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.279	0.317	0.366	0.401	0.424	0.399	0.379	0.346	0.306	0.269	0.253		
	taud		2.368	2.307	2.232	2.132	2.078	2.101	2.240	2.273	2.311	2.365	2.385	2.359		
	Ebn at Noon		280	290	290	283	275	269	274	276	277	275	271	268		
	Edh at Noon		24	31	38	46	50	49	43	40	35	29	23	22		
(44) All-Sky Solar Radiation	RadAvg		695	1035	1569	1965	2244	2208	2104	1941	1627	1204	799	592		
	RadStd		66	101	61	90	95	119	109	66	59	43	46	62		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (3 neighbors)	+0.59	N/A	N/A	N/A	-1.53	-1.48	N/A	N/A	N/A	N/A

Nomenclature: See separate page